

# Project Report: Online Survey & Feedback System

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## 1. Project Overview & Vision

The Online Survey & Feedback System is a full-stack web application designed to simplify survey creation, distribution, and analysis. It empowers administrators or organizations to create customized surveys, collect participant feedback, and visualize results in real time. Unlike traditional survey tools that provide limited analytics, this system leverages dynamic form-building, cloud deployment, and AI-ready architecture for future scalability.

For participants, the system ensures ease of use, allowing them to quickly access surveys, submit responses, and view acknowledgment. For administrators, the platform provides advanced result dashboards with visual statistics, response aggregation, and filtering options.

The vision is to create a scalable, secure, and intelligent survey platform that can serve educational institutions, businesses, and organizations for gathering structured feedback and insights.

## 2. Technical Foundation

The system is built on a modern **full-stack architecture**, ensuring modularity, scalability, and performance.

### a) **Frontend:** React

- Provides an interactive and responsive user interface.
- Dynamic form builder for creating surveys.
- Data visualization using chart libraries.

### b) **Backend:** Spring Boot

- Exposes RESTful APIs for survey creation, response handling, and analytics.
- Spring Security for authentication & authorization (role-based: Admin, Participant).

### c) **Database:** MongoDB

- Stores survey metadata, participant responses, and aggregated results.
- Flexible schema to support dynamic survey questions.

d) **ORM Layer:** Hibernate with Spring Data JPA for smooth DB operations.

e) **Security:**

- Spring Security with JWT for token-based authentication.
- Role-based access control.

f) **Cloud Deployment:**

- Deployed on AWS/GCP/Azure (containerized using Docker).
- Load balancing for high availability.

g) **Caching:**

- Redis caching for faster response retrieval and analytics aggregation.

This stack ensures a **robust, secure, and scalable** architecture.

### 3.Core Modules

#### 3.01 Survey Builder (Admin Module)

- Create, edit, and manage surveys.
- Dynamic form builder supporting multiple question types (MCQ, text, rating, dropdown).
- Publish and unpublish surveys.
- Set survey availability time and participant access rights.

#### 3.02 Participant Portal

- Access available surveys through unique links or login.
- Submit responses securely.
- Option to provide feedback anonymously.
- View acknowledgment after submission.

#### 3.03 Result Dashboard

- Aggregate and analyze participant responses.
- Visual statistics (pie charts, bar graphs, trends).
- Filter responses by date, demographic, or question type.
- Export reports in CSV/PDF format.

## 4. Key Evaluation Aspects

### 1. Dynamic Survey Creation

- Flexibility in building customizable surveys.
- Supports various question formats.

### 2. Real-Time Result Aggregation

- Aggregates responses instantly.
- Generates interactive visual statistics.

### 3. Secure Role-Based Access

- Admin vs Participant roles.
- Authentication via JWT and Spring Security.

### 4. Scalability & Performance

- Caching for fast analytics.
- Cloud deployment for horizontal scaling.

## 5. How the Tech Stack Supports This

- **React:** Provides modular, reusable UI components and dynamic survey builder.
- **Spring Boot:** Robust backend framework with built-in support for REST APIs and security.
- **MongoDB:** Schema flexibility for storing dynamic survey data.
- **Spring Security + JWT:** Secure role-based authentication.
- **Redis Caching:** Improves response retrieval speed.
- **Cloud:** Ensures high availability and load balancing for large-scale usage.

## 6. System Design

### High-Level Design (HLD)

At the highest level, the system consists of:

- **Frontend (React)** → Interacts with users (Admins & Participants).
- **Backend (Spring Boot)** → Handles API requests, business logic, and security.
- **Database (MongoDB)** → Stores surveys and responses.
- **Caching Layer (Redis)** → Speeds up analytics.

- **Cloud Infrastructure** → Ensures scalability and availability.

### **Low-Level Design (LLD)**

- Admin creates survey → Backend stores schema in MongoDB.
- Participant submits response → Backend validates and stores response.
- Dashboard queries results → Cached data retrieved for speed.
- Analytics → Visual charts rendered on frontend using React libraries.

## **7.0 Future Scope & Enhancements**

- **AI-Powered Insights:** Sentiment analysis of open-text feedback.
- **Adaptive Surveys:** Dynamic branching based on participant answers.
- **Integration with LMS/ERP:** Plug-in support for institutions.
- **Gamified Surveys:** Engage participants using rewards and points.
- **Mobile App:** Dedicated iOS/Android app for broader accessibility.
- **Multi-Language Support:** Surveys in regional languages.